

SIERRA NELSON

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EDUCATION

Florida Agricultural & Mechanical University - FAMU-FSU College of Engineering

Tallahassee, FL

Bachelor of Science in Mechanical Engineering

May 2027

GPA: 3.38/4.00 | **Dean's List:** Fall 2022, Fall 2024, Fall 2025 | **NSF- STEM Scholar:** 2023-2024

Honor Roll: Spring 2023, Fall 2023, Spring 2024, Spring 2025, Spring 2026

Relevant Coursework: Mechanical Systems, System Dynamics & Vibrations, Engineering Design Methods, Mechatronics, Materials Science, Thermal-Fluids, Control Systems, Advanced Dynamics

TECHNICAL SKILLS

CAD & Design: SolidWorks, Siemens NX, AutoCAD (Certified), Tinkercad, Shapr3D

Additive Manufacturing: FDM 3D Printing, Bambu Lab A1, Bambu Studio, Ultimaker Cura, slicing, rapid prototyping

Mechanical Engineering: Mechanical Design, FEM, Design Optimization, Mechatronics, Prototype Testing

Technical Computing: MATLAB, working knowledge of Python (NumPy, pymoo) and C++

ENGINEERING EXPERIENCE

Print Pro 3D Designs | Tallahassee, FL

Founder & Additive Manufacturing Designer

November 2024 - Present

- Founded and grew a profitable 3D-printing business, converting customer ideas into custom physical products from concept through delivery.
- Create and refine CAD models in Shapr3D and Tinkercad, prepare parts in Bambu Studio, and operate a Bambu Lab A1 for repeatable FDM production.
- Manage the complete build workflow, including requirements, slicing, printer setup, troubleshooting, quality inspection, client communication, pricing, and scheduling.

Penn State Applied Research Laboratory | State College, PA

May 2024 - July 2024

Research & Design Engineer - ONR Pipeline Program

- Optimized airfoil material and geometry parameters to reduce vibration using a Python-based genetic algorithm and engineering design constraints.
- Applied finite element methods to assemble mass and stiffness matrices, calculate eigenvalues and natural frequencies, and compare design alternatives.
- Presented the final technical solution and received the 2024 Best Project Award for the ONR Pipeline Program.

Penn State Applied Research Laboratory | State College, PA

Mechanical Design Intern

May 2023 - August 2023

- Designed and prototyped a buoyancy-control system for an autonomous robotic fish in collaboration with a graduate researcher.
- Applied fluid-mechanics calculations to determine required water volume and support stable underwater operation.
- Evaluated simulations and test results to refine components through iterative design and hands-on prototyping.

SELECTED DESIGN PROJECTS

- Thermal Characterization of 3D-Printed Metals - Senior Design (in progress):** Developing an experimental approach to evaluate how additive-manufacturing variables affect thermal behavior.
- Bluetooth-Enabled Oven - EESI:** Collaborated on an industry-sponsored concept and designed space-efficient mechanical solutions integrating oven hardware with a mobile viewing application.
- Mechanical CAD & Prototyping:** Redesigned an Iron Man arc reactor in SolidWorks and designed and built a functional mechanical can crusher.

LEADERSHIP

Tau Beta Sigma National Honorary Band Sorority, Inc. - Vice President of Special Projects

FAMU Marching '100' - Rank Sergeant